

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY (Deemed to be University)
M.Tech/ Ph.D Autumn Semester End Sem. Exam Dec. 2022

Subject : Scientific Computing

Course Code : AM 6060

Duration : 03 Hrs.

Max Marks : 50

- a. Marks are indicated against each question.
b. Before attempting please ensure that you have the correct question paper.

1. a) Find a linear mapping $F : R^3 \rightarrow R^3$ whose image is spanned by $(1,2,3)$ and $(4,5,6)$. [3]
b) Find a linear mapping $G : R^4 \rightarrow R^3$ whose kernel is spanned by $(1,2,3,4)$ and $(0,1,1,1)$. [3]
c) Find the homogeneous system whose solution space is spanned by $(1,1,2,1,1)$, $(1,2,1,4,3)$, $(3,5,4,9,7)$. [4]
2. a) Show that the Moore Penrose inverse is unique. Find the Moore Penrose inverse for $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$ using QR decomposition. [10]
3. a) Discuss Overfitting and Underfitting for system of equations with examples. Also give the solution by writing appropriate Pseudo inverse in both the cases. [4]
b) Find the largest eigen value and the corresponding eigen vector using power method for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ [6]
4. a) Give the algorithm for solving system of equations using LU decomposition. [5]
b) Discuss Gram Schmidt orthonormalization procedure. [5]
5. a) Using 3 point Gauss quadrature, evaluate $\int_0^1 \frac{dr}{1+r^2}$ [5]
b) Discuss PCA in detail and give its application in Data mining with an example. [5]